FROM CURRENT PROGRESS TO FINAL SUBMISSION

LG TV conversion
Project execution plan

A practical companion to the reviewed rationale. English working version | 25 October 2026

The project we are taking forward

Compare LG and Samsung TV view-to-purchase conversion, accounting for initial price and time. Use the observed journey to identify where further investigation or a retailer-supported test would be most useful.

Start from the work already done

The fresh raw-data audit reproduces the cohort and baseline Logit and adds targeted method checks. Product concentration changes the estimated gap materially. The group should now reproduce this work in its own notebooks and develop a defensible interpretation. The direction remains subject to group agreement and instructor feedback. A smaller or unstable gap is an informative result.

Official milestone, Milan time	What the group brings or submits
31 Oct 2026, 23:59	Interim PDF, maximum five slides.
5 Nov, 12:00-13:30, Aula 34	Tutorial I: progress, definitions and decisions needing feedback.
16 Nov, 16:30-18:00, Aula 34	Tutorial II: findings, robustness and provisional recommendation.
25 Nov, 23:59	Final slides, report and code archive on Blackboard.
26 Nov, 10:15-13:30, Aula 34	Class presentation: 8 minutes plus 3 minutes of Q&A.
30 Nov, 16:30-18:00	LG manager presentation for shortlisted teams; room TBD.

How to use this plan

Steps 1-3 prepare the interim submission. Steps 4-8 develop and challenge the evidence. Steps 9-12 turn that evidence into the assessed deliverables. Dates within those steps are suggested working targets; only the milestone table comes from the official guidelines. Check Blackboard for course amendments.

Assign a lead and a reviewer for each output: data/cohort, modelling, interpretation and communication. Everyone should understand the research question and main results. Keep decisions in notebook Markdown or the README; no additional setup framework is necessary.

Sources: 30420_Group_Project_Guidelines.pdf, pp. 1, 3-5 [G]; supplied project rationale [R]; LG_TV_Work_Handoff_EN_v3.md [H]. This is a work plan, not final report text. Full source map: page 8.

STEPS 1-3 | SUGGESTED WINDOW: 25-31 OCTOBER

Prepare the interim slides

Step 1. Agree the question and the managerial decision

Read sections 1-6 of the rationale together. Confirm TV, LG versus Samsung, October-November 2019 and the single behavioural dataset. Explain why the benchmark is useful while acknowledging that similar prices do not establish comparable specifications. Record the group number, member names and upload responsibility.

Output: one agreed paragraph containing the business problem, primary RQ and decision. The question concerns an adjusted conversion comparison; the decision concerns where to investigate. Do not promise to prove defective pages, checkout friction or effective discounts.

Step 2. Reproduce the existing cohort and feasibility evidence

Load **electronics_market_behavior.csv** from the private sibling data folder. Keep the raw file unchanged and notebook paths relative. Use user_id x session_id x product_id, at least one view, and a purchase strictly after the first view within that key. Retain initial price and time. Explain every exclusion in Markdown.

Reference check from existing work	LG	Samsung
Eligible consideration rows	22,213	52,499
Purchase outcomes	505	1,645
View-to-purchase	2.27%	3.13%
Distinct products	105	91

Reconcile discrepancies before copying numbers into slides. Check filters, user keys, strict timestamps and counting rules first. Preserve the reported 1,247 duplicate-event issue and four session IDs shared across users in the audit. The counts above were independently reproduced from the raw CSV in the current audit [A]. The group should reconcile its own notebook to them.

Step 3. Build the five-slide proposal

1. Brand and context: group and members, brief LG introduction, TV scope and choice rationale. **2. Problem:** the preliminary conversion difference and the investigation decision. **3. RQ:** primary adjusted comparison, supporting journey diagnostic and scope. **4. Data:** file, period, unit, counts and historical retailer-proxy limits. **5. Approach:** journey KPIs, binary Logit and the main checks still open.

Export **30420_Interim_Group[N].pdf**, replacing [N]. One member uploads to Blackboard by **31 October, 23:59 Milan time** and verifies the submission. The five-slide allocation is a suggestion; the maximum and required content are official. A completed model or definitive intervention is not required. Interim content is not graded, but non-submission incurs a deduction. [G, pp. 3-4]

Ready to move on when...

All five slides tell the same story, counts and definitions agree, and everyone can distinguish what is observed from what is still a hypothesis.

STEPS 4-5 | SUGGESTED WINDOW: 1-5 NOVEMBER

Make the comparison defensible

Step 4. Document exactly what enters the analysis

Create an audit from raw events to TV events, then LG/Samsung events, then eligible consideration rows. Record date conversion, missing values, invalid prices, duplicates, brand/product consistency and exclusions. Confirm that each analytical key is unique and Y is binary. The reference cohort contains 74,712 rows and 2,150 purchases; these are not 74,712 independent people.

Write the KPI definitions beside the code. View-to-cart requires a cart after the first view. Cart-to-purchase requires a purchase after an eligible cart and uses only rows with that cart in its denominator. Count purchases without a qualifying cart separately. Do not force the two intermediate rates to multiply to overall conversion.

Output: a reproducible cohort audit and a short measurement note that another member can use to rebuild the table. A zero outcome means no qualifying purchase observed in this unit, not no purchase elsewhere or later.

Step 5. Check whether the two brands overlap enough

Plot full initial-price distributions by brand and report a few quantiles, not only medians. Inspect the number and share of observations and purchases by product, user and month. This shows whether the comparison relies on non-overlapping prices or a small number of highly active products or people.

If a common-price range or exclusion is justified, state the rule before interpreting its result and apply it symmetrically. Preserve the full cohort as the baseline and label any restricted sample. Report changed counts and the narrower population to which the comparison applies. Product specifications remain unobserved even inside price overlap.

Starting evidence: the top five products account for 42.3% of LG rows and 52.4% of Samsung rows. **Output:** a price-overlap chart, concentration summary and justified sensitivity samples. Do not silently replace the cohort or keep using the old denominator after a restriction.

Tutorial I: 5 November, 12:00-13:30, Aula 34

Bring the RQ, cohort definition, reproduced summary, price-overlap evidence and proposed M1/M2. Ask for feedback on the user-session-product unit, comparability and inference with repeated users and products. Explain what has changed since the interim and what comes next. Record the feedback and any resulting scope changes immediately afterwards.

Ready to move on when...

The group can explain what each row and rate means, identify the limits of comparability, and justify its modelling sample. Major unresolved choices have been raised with the instructor.

Course links: Session3_CustomerJourney.pdf; Session5_Computing_Visualizing_KPIs_python.pdf; session6_kpi_case.ipynb. Audit and sensitivity choices are project recommendations, not additional official deliverables.

STEPS 6-7 | SUGGESTED WINDOW: 6-10 NOVEMBER

Estimate and interpret the model

Step 6. Fit a small, transparent model sequence

Follow Session 12 and the Session 15 response-model notebook. Use Samsung as reference, define LG = 1 for LG, and retain the same sample for the first two models. Log initial price requires strictly positive prices.

Main specifications

M1: logit[P(Y=1)] = b0 + b1 x LG
M2: logit[P(Y=1)] = b0 + b1 x LG + b2 x ln(initial price) + b3 x November

Keep only predictors known at the first view. Later carts, final spending and total-session events would use future information. Product fixed effects cannot identify a separate brand coefficient when each product belongs to one brand. Keep convergence warnings visible and check sparse cells or separation before interpreting coefficients.

The baseline uses user-clustered standard errors. The audit also checks product clustering separately and finds wider uncertainty. Neither separate calculation jointly handles both structures; discuss final inference with the instructor. Fit statistics do not address dependence or omitted variables. Do not transfer OLS adjusted R-squared or F-tests into a Logit table; label pseudo-R-squared correctly.

Step 7. Translate the comparison into probabilities

Report the LG odds ratio with its interval as a technical result, then make the adjusted probability difference the managerial summary. For each eligible row, predict once with LG and once with Samsung, keeping its price and month fixed. Average each set of predictions and the paired differences. Use unrounded values to calculate the percentage-point gap.

Reproduce the probability-gap interval, then agree the final dependence treatment. The audit obtains -0.827 points, with an approximate user-clustered 95% delta-method interval of [-1.107, -0.546]. Product clustering separately widens it to [-1.475, -0.179]. These conditional intervals do not cover omitted-variable bias. The reference probabilities remain 2.295% and 3.121%.

Output: a compact M1/M2 table with N, purchases, controls, uncertainty treatment and the brand contrast; a probability plot with intervals; and a short explanation of why raw and adjusted comparisons differ or remain similar. Keep the standardisation population explicit, especially for restricted samples.

Ready to move on when...

Another member can reproduce both models, explain odds versus probabilities, and interpret the main contrast without claiming that the brand or price coefficient is causal.

Direct course grounding: Session12_Logit.pdf, pp. 7-22; session15_response_models.ipynb, sections 1-3; Session7_OLS_Extention_Application.pdf, pp. 10-13. Dependence-aware inference choices should be confirmed with the instructor.

STEP 8 | SUGGESTED WINDOW: 10-16 NOVEMBER

Challenge the result before acting

Step 8. Run checks tied to concrete concerns

Concern	Check and output
Price and product mix	Reproduce the central-price restriction: N=71,232, gap -0.885 pp. Removing each brand's most-viewed product gives N=62,883, gap -0.431 pp. Investigate the changed mix.
Timing and observation window	Test defensible rules for equal timestamps and end-of-window exposure; compare months or finer time controls.
Repeated observations	Document users and products contributing multiple rows. Agree how uncertainty should reflect dependence.
Functional form	Retain LPM as a supporting check: gap -0.822 pp versus Logit -0.827 pp; no observed fitted LPM values outside [0,1].
Different price associations	The audited LG x log(price) interaction adds little fit and barely changes the average gap. Keep the simpler model unless a new substantive reason emerges.

Keep a sensitivity log: concern, rule, changed sample, estimated contrast and interpretation. A changed sample answers a somewhat different question, so separate changes in population from changes in specification. Report fragility openly. Do not add interactions or replace the benchmark merely to recover significance.

Choose extensions with a stopping rule

The core remains journey KPIs plus Logit. The audited LPM supports the same average comparison; the interaction adds little. Concentration and dependence checks now deserve more effort than further techniques. Stop adding models when they no longer resolve a stated uncertainty or affect the managerial interpretation.

K-means/RFM would change the focus to targeting; PCA requires defensible attribute measurements; multinomial Logit requires a clearly defined choice outcome and alternatives. Pricing elasticity and promotion models need data structures absent here. The course review in the companion guide explains these decisions; covering every taught method is not a project requirement.

Tutorial II: 16 November, 16:30-18:00, Aula 34

Bring the main probability contrast, its uncertainty, the most consequential sensitivity results and a provisional recommendation with limitations. Explain progress since Tutorial I. Ask which interpretation the evidence supports and which remaining check would change the decision. Record the feedback and close the analytical scope unless a material problem remains.

Ready to move on when...

The group knows whether its main conclusion is stable, limited to a subset or inconclusive, and can show why. Unresolved limitations remain visible rather than being disguised by more models.

Course links: Session 7 (interactions and model comparison); Session 12 (LPM and Logit); Sessions 8-11 (segmentation/PCA); Sessions 16-17 (pricing). Sensitivity checks are proposed analytical safeguards.

STEP 9 | SUGGESTED WINDOW: 17-19 NOVEMBER

Turn findings into a decision

Step 9. Write the evidence-to-action argument together

Start with the result that survives the checks: direction, magnitude, uncertainty, population and limits. Then separate the observed pattern from possible explanations. Conclude with the next action and the evidence needed to justify it. The current historical proxy supports prioritising an investigation more directly than implementing a price or checkout change.

If the evidence shows...	A defensible next decision
A persistent view-to-cart difference	Prioritise a product-information and commercial-terms audit. Request specifications, stock and traffic-source evidence before proposing a page-content test.
A difference among eligible cart users	Request delivery, payment, stock and error information. Investigate with the retailer; selected cart users do not identify checkout friction.
A gap concentrated in a subset	Investigate that assortment or price range. Explain the narrower target and avoid a catalogue-wide recommendation.
A small, unstable or inconclusive gap	Recommend better evidence or no immediate intervention based on this study. State what uncertainty prevents action.

Make a future test specific without inventing its economics

If the evidence supports a test proposal, specify one change, the eligible audience, channel, responsible party, allocation unit, comparison and primary purchase outcome. A retailer would need to agree any change to its website. State a consistent observation window and any guardrails requiring new data, such as returns or margin.

Do not invent a sample size, duration, budget, expected lift or ROI. Those require current traffic, a worthwhile effect, costs and an implementable design. Explain if the future test uses visitor-level outcomes while the historical study uses product considerations. This is a proposal for future evidence, not an experiment already conducted.

Keep limitations connected to the decision

Unobserved specifications, availability, promotions, delivery and purchase intent may explain the adjusted gap. The retailer, country and currency are unidentified, and the period is 2019. State which conclusion these gaps weaken and what additional evidence would resolve them. Do not treat the dataset as current LG performance or the entire TV market.

Ready to move on when...

The recommendation names an action, a responsible party and the next evidence required. It remains defensible if the gap is weaker than the initial result.

Course links: value and KPIs (Sessions 2, 5); touchpoint ownership (Session 3); pricing and contribution limits (Sessions 16-17). Managerial directions build on [R], sections 18-21, and [H].

STEPS 10-12 | SUGGESTED WINDOW: 19-26 NOVEMBER

Prepare, verify and submit

Step 10. Assemble the final communication

The group writes the report using the official logic: managerial problem, RQ, data, methods, findings, interpretation, recommendation, limitations and next evidence. Use **maximum 12 pages including tables and figures, 12-point font, 1.5 spacing and APA style**; the reference list is outside the limit. The course permits AI for coding, debugging and grammar checking, not drafting the final report; include the required AI-use disclosure. [G, p. 5]

Build final slides around the same argument. Use readable charts and a few material results; keep technical detail available for questions. The guidelines specify an eight-minute final presentation, not a final slide-count limit. Rehearse with a timer and prepare answers on the unit, comparator, uncertainty, omitted variables and recommendation.

Step 11. Test reproducibility and reconcile every number

Have a member run the workflow from a fresh kernel using their private dataset copy and the documented relative paths. The existing sibling **data/01_LG_Electronics_Appliances/** structure can remain. Include all code/notebooks that clean, analyse and generate outputs, with run order and required libraries explained in the README. Intermediate files must use exactly the names expected downstream.

Create the code archive from the necessary reproducible workflow. Keep private raw datasets outside Git and document the required course input file. Check that tables, plots, slide text and report text use the same cohort and model version. Label raw and adjusted gaps separately; avoid personal absolute paths and unintended raw-row exports.

Step 12. Submit and present the frozen version

One member uploads the following to the correct Blackboard assignments by **25 November 2026, 23:59 Milan time**:

30420_Slides_Group[N].pdf
30420_Report_Group[N].pdf
30420_Files_Group[N].zip or **.rar**

Replace [N], open the exported PDFs, inspect the archive and verify the saved submissions. Deadlines are strict and late files are not accepted through another channel. The uploaded slides are the slides presented. On **26 November**, deliver eight minutes plus three minutes of Q&A; presenters may not read paper or phone notes. [G, pp. 3-5]

If shortlisted, prepare for the LG manager session on **30 November, 16:30-18:00**, following course instructions. Complete the individual peer evaluation through the Qualtrics link posted on Blackboard after the final presentations and before scores are released.

Ready to move on when...

The submitted files open, the analysis reruns, the figures agree, and the group can explain the result and recommendation within the presentation time.

REFERENCE PAGE

Course map and working sources

Teaching material	Role in this project
Session2_ValueFramework_KPIs_260904.pdf; Session3_CustomerJourney.pdf	Connect the commercial decision to an observable KPI; distinguish the partial recorded journey and touchpoint ownership.
Session5_Computing_Visualizing_KPIs_python.pdf; session6_kpi_case.ipynb	Compute rates with explicit denominators and communicate comparisons.
Session4_LinearRegression.pdf; Session7_OLS_Extention_Application.pdf; session7_ols_application-2.ipynb	Controls, interaction logic and model comparison. OLS-specific diagnostics do not automatically transfer to Logit.
Session12_Logit.pdf; session15_response_models.ipynb	Primary method: binary Logit, odds ratios, probability interpretation. LPM is optional; multinomial models address a different outcome.
Session8_STDP_Segmentation.pdf; session9_segmentation_case.ipynb	RFM and k-means are alternatives for a targeting question, not required additions to this brand comparison.
Session10_Positioning_Perceptual_Map.pdf; session11_positioning_perceptual_map-2.ipynb	PCA and perceptual mapping require suitable attributes/perceptions not present in this TV event sample.
Session16_Pricing_Promotion.pdf; session17_pricing_case.ipynb	Elasticities, promotions and contribution analysis clarify why the present price coefficient cannot support a discount or profit claim.

Project sources and evidence status

[G] 30420_Group_Project_Guidelines.pdf, pp. 1-5: official requirements and supplied dates.
[R] LG_TV_Project_Rationale_and_Plan_EN.pdf: preferred eight-page rationale and existing evidence.
[H] LG_TV_Work_Handoff_EN_v3.md, 25 October 2026: progress record, corrected counts and open decisions.
[D] DATA_DICTIONARY.md in the LG course dataset folder: data structure and limitations.

[A] denotes the fresh audit of both raw CSVs on 25 October 2026, including independent cohort construction, Newton maximum-likelihood Logit, least-squares LPM and the specified sensitivity fits. Baseline results reconcile with [R/H]. Intervals use cluster sandwich covariance and the delta method, holding standardisation covariates fixed. This is not a rerun of the earlier scripts; timing and other robustness work remain open. Recommended checks are not new official requirements.

Working documents belong with the group notebooks and observations. Course datasets remain in each member's private local folder. The separate reviewed rationale explains why the project is framed this way; this plan explains what to do, what to produce and when to move on.